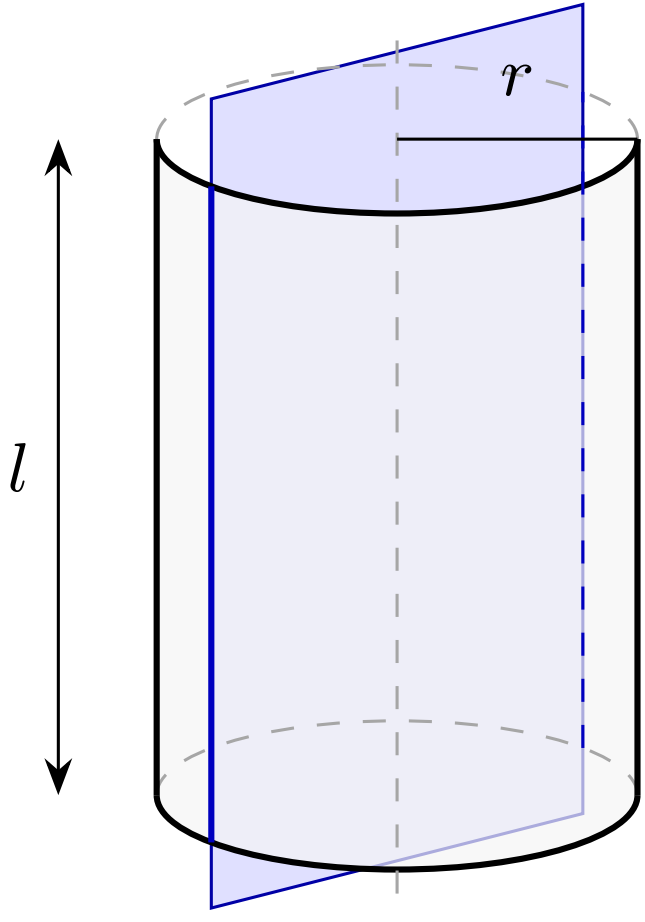
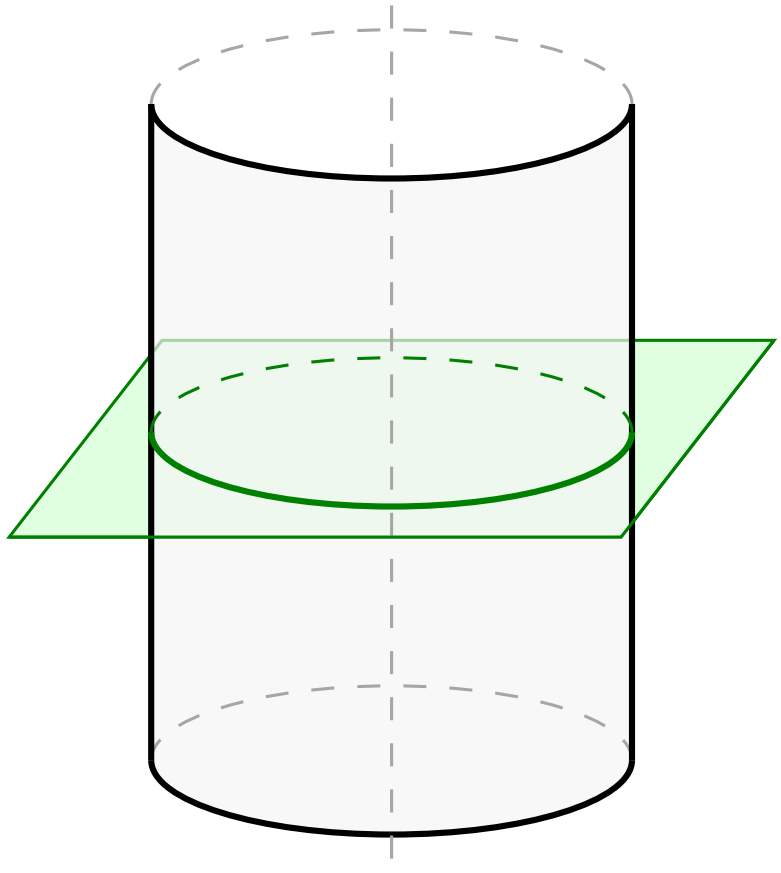
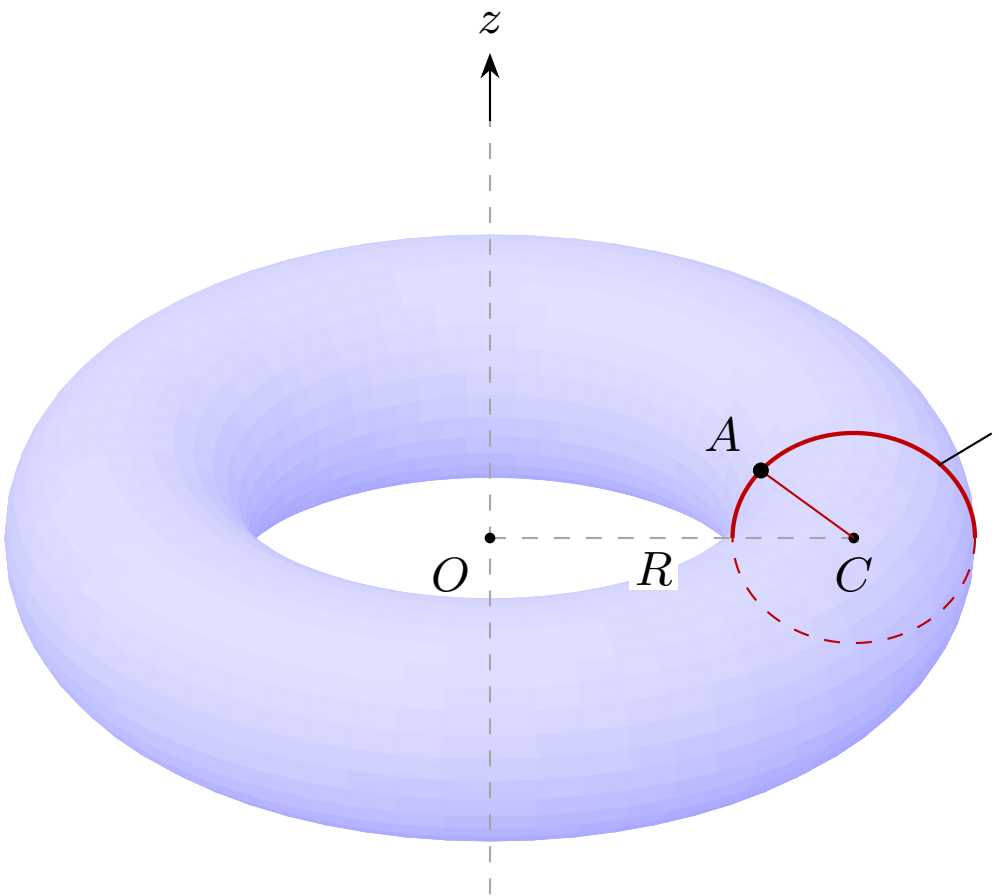
纵切面：含中轴线



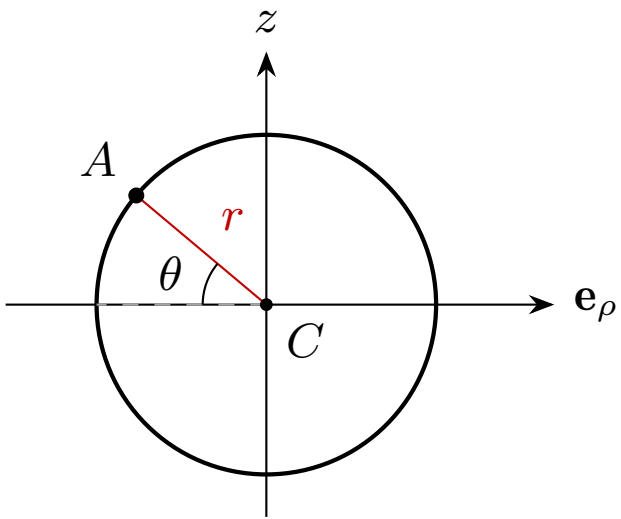
横切面：垂直中轴线





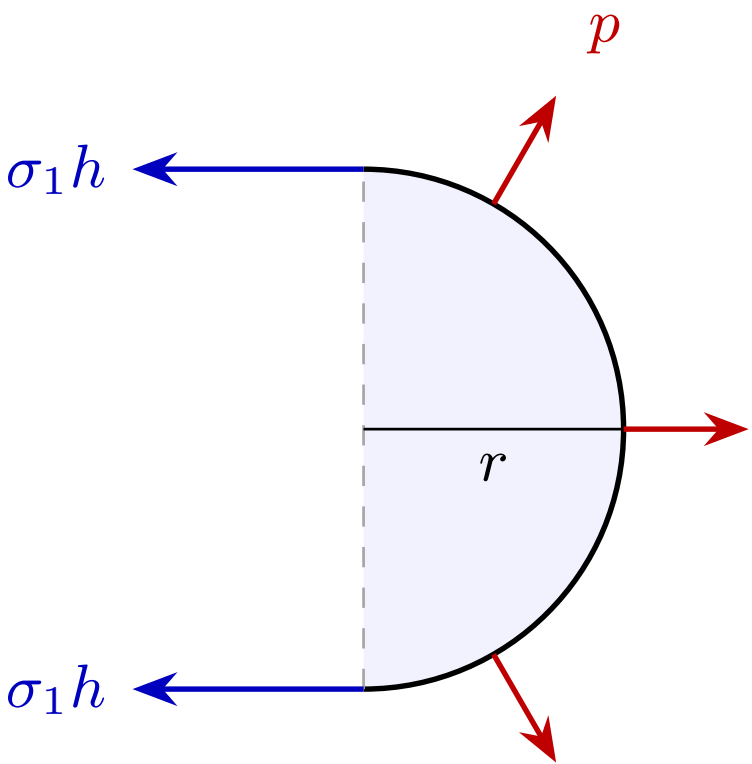
子午横截面

子午截面；内侧在左



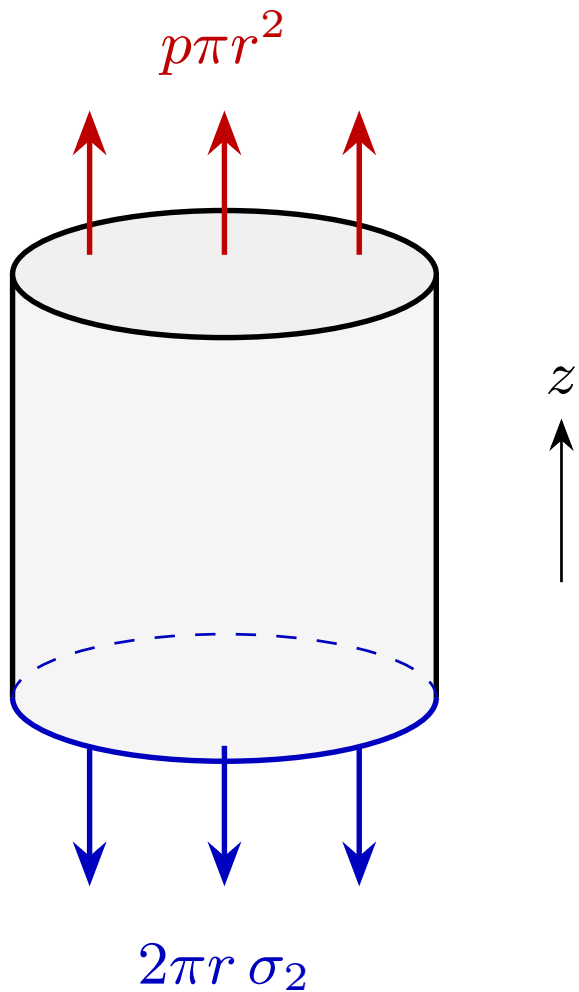
$$s = R - r\cos\theta, \quad z = r\sin\theta$$

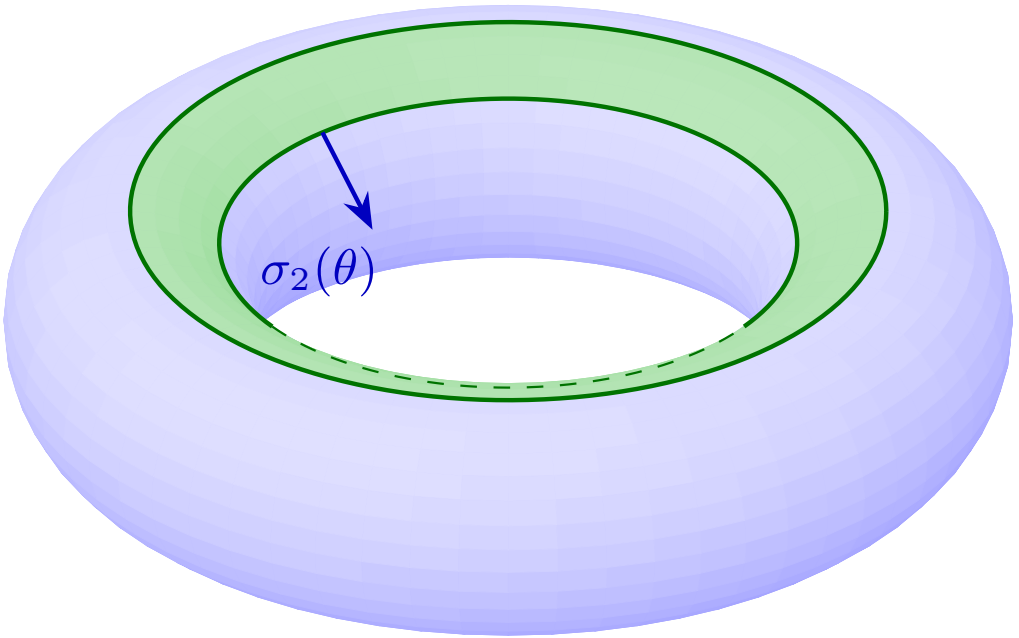
半筒横截面；轴向长 h



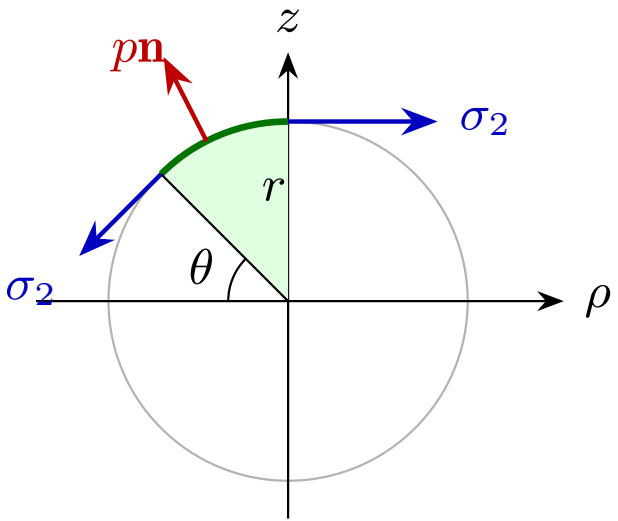
$$F_{p,x} = 2prh$$

闭端筒段：轴向平衡

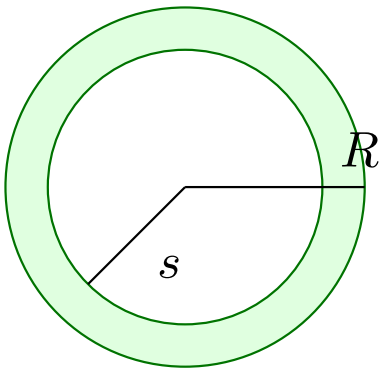




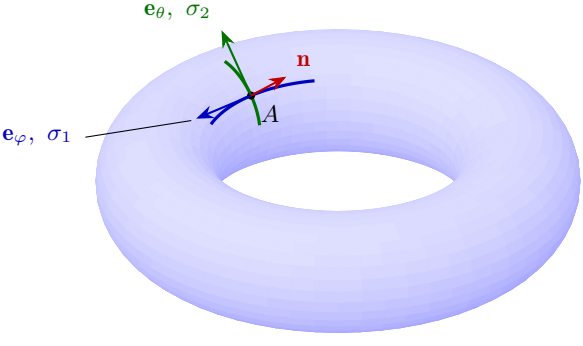
内侧上膜带；两边界为平行圈



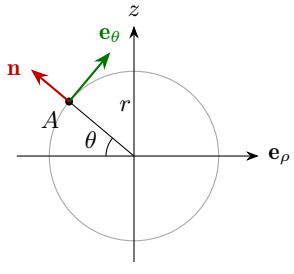
顶部子午力的 z 分量为零



投影面积 $\pi(R^2 - s^2)$

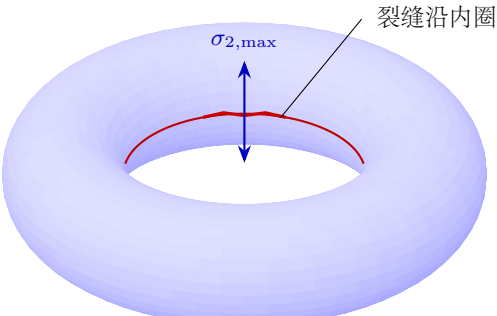


两个力均在膜的切平面内



$$\kappa_\theta = \frac{1}{r}$$

$$\kappa_\varphi = -\frac{\cos\theta}{R - r\cos\theta}$$



最大拉力沿子午方向

示意取 $R = 3r$

